

CHARACTERIZATION OF A DIGITAL MUSIC STREAMING UNIVERSE

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1 | Files included in the ZIP archive

1. Images of ER diagrams in SVG format: 3 diagrams – full, one containing only entity types and attributes and the third containing only relationships (i.e., without the attributes of entity types). The explanation of the translation from requirements to ER model is in this document;
2. Image of the relational model diagram in SVG format; The explanation of the translation from ER model to the relational model is in this document;
3. SQL file containing the schema and data of the database;
4. SQL file containing examples of queries to the database;
5. File with Python code;
6. Data files (CSV files) that were used by the Python script to populate the database.

2 | DB Stored Information and Requirements

In this project, a fictional universe of a company providing a music streaming service was characterized. The database (DB) stores information related to:

- User data, including identification (UserID), name, email, registration date, account length and the user's chosen plan type (referring to the subscription or pricing plan with options such as free, premium, family and student);
- Songs, including identification (SongID), title, artists (potentially more than one), number of artists, duration, and optionally, the song's genre. Additionally, it stores user engagement data such as the number of total plays and total downloads;
- Playlists, including identification (PlaylistID), title, privacy setting (describing the privacy settings of each playlist, whether public or private) and optionally, description;
- Devices, including identification (DeviceID), type, model, and total and free storage capacity. In this context, the "Devices" mentioned in the database include various electronic devices that users can use to access and enjoy the streaming platform – the options for the device type are: smartphones, tablets and computers.

Additional recorded information includes the creation date of each playlist, the user subscription date to a specific playlist, the play date of a playlist on a given device, the date of the 'like' given by the user to a song, and the user connection date to a specific device.

Additionally, in terms of relationships it is established that:

1. Each user can create several playlists, but each playlist has only one creator (1-to-many relationship);
2. Each user is allowed to subscribe to various playlists, and each playlist can have several subscribers (many-to-many relationship);
3. Each playlist can be played on various devices, and each device can play several playlists (many-to-many relationship);
4. Each user is allowed to like multiple songs, and each song can have several likes from different users (many-to-many relationship);
5. One song can belong to several playlists, and one playlist can have multiple songs (many-to-many relationship). In this universe, there are empty playlists, without songs;
6. One user can connect to multiple devices, and each device can be connected to several users (many-to-many relationship).

With the exception of the first relationship, both entities in each relationship exhibit partial participation. For instance, considering statement number 2 above: some users might not subscribe to any playlists (partial participation), and some playlists might not be subscribed to by any users (partial participation).

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Regarding statement number 1: Some users may not create playlists (partial participation), but every playlist must have one creator (total participation).

In summary, the primary purpose of this database is to organize, manage, and enhance interactions with various components, including music tracks, playlists, user profiles, and connected devices. It handles a vast array of songs, ensuring a diverse and inclusive music library for users. The system is designed to be compatible with a wide range of devices, from smartphones to smart speakers, offering a versatile and accessible music streaming experience. Its scope extends to a global user base, embracing a broad audience with diverse musical tastes.

3 | ER model

As explained, the ER model is presented in the ZIP archive.

In the following table (Table 1), it is explained the type of attributes presented in the ER model.

Table 1. Entities and types of attributes in the ER model.

Entity	Attribute	Type
USER	UserID	Key attribute
	Name	Normal
	Email	Normal
	RegistrationDate	Normal
	AccountLength	Derived
	PlanType	Normal
SONG	SongID	Key attribute
	Title	Normal
	Artist	Multi-valued
	NumArtists	Derived
	Genre	Optional
	Duration	Normal
	UserEngagement: - Plays; - Downloads	UserEngagement: Composite Plays and downloads: normal
PLAYLIST	PlaylistID	Key attribute
	Title	Normal
	Description	Optional
	PrivacySetting	Normal
DEVICE	DeviceID	Key attribute
	Type	Normal
	Model	Normal
	StorageCapacity: - Total; - Free	StorageCapacity: Composite Total and Free: normal

In terms of relationships, the explanations are presented in Table 2. This is the translation from requirements shown in page 1 to the ER model.

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Table 2. Relationships, entity participation, cardinality, and its attributes in the ER model.

Relationships	Participation	Cardinality	Attributes
CREATOR_OF (USER, PLAYLIST)	Partial > total	1 : N	CreationDate
SUBSCRIBE_TO (USER, PLAYLIST)	Partial > Partial	M : N	SubsDate
PLAYED_ON (PLAYLIST, DEVICE)	Partial > Partial	M : N	PlayDate
LIKES_THE (USER, SONG)	Partial > Partial	M : N	LikeDate
BELONGS_TO (SONG, PLAYLIST)	Partial > Partial	M : N	-
CONNECTED_TO (USER, DEVICE)	Partial > Partial	M : N	ConnectDate

Key attributes are those that uniquely identify each entity within a database:

- UserID serves as the primary key for the USER entity. It uniquely identifies each user within the system;
- SongID is the primary key for the SONG entity. It uniquely identifies each song within the music streaming service;
- PlaylistID acts as the primary key for the PLAYLIST entity. It uniquely identifies each playlist created on the platform;
- DeviceID serves as the primary key for the DEVICE entity. It uniquely identifies each device registered in the system.

Furthermore:

- AccountLength is the period of time during which the account has been active and is derived from the RegistrationDate (calculated as the difference between the current date and the registration date) and NumArtists is a derived attribute from Artist;
- UserEngagement is an attribute composed of the sub-attributes Plays and Downloads. StorageCapacity is an attribute composed of the sub-attributes Total and Free;
- The attributes Genre (in SONG) and Description (in PLAYLIST) are optional and can take a NULL value (be undefined);
- The attribute Artist is multi-valued: it is considered that a song can feature the participation of multiple artists.

2.1 Description of entity types using the textual syntax of the ER model:

USER(UserID, Name, Email, RegistrationData, [AccountLength], PlanType)

SONG(SongID, Title, {Artist}, [NumArtists], Genre?, Duration, UserEngagement(Plays,Downloads))

PLAYLIST(PlaylistID, Title, Description?, PrivacySetting)

DEVICE(DeviceID, Type, Model, StorageCapacity(Total,Free))

Regarding the relationships:

CREATOR_OF(USER,PLAYLIST, CreationDate)

SUBSCRIBE_TO(USER,PLAYLIST, SubsDate)

PLAYED_ON(PLAYLIST,DEVICE, PlayDate)

LIKES_THE(USER,SONG, LikeDate)

BELONGS_TO(SONG,PLAYLIST)

CONNECTED_TO(USER,DEVICE, ConnectDate)

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The convention is the following:

Key attributes written as: Attribute

Optional attributes written as: Attribute ?

Derived attributes written as: [Attribute]

Multi-valued attributes written as: {Attribute}

Composite attributes written as: Attribute(Sub-Attribute1, Sub-Attribute2)

REL(E1,E2, Attribute)

ENTITY(A1,...,An)

2.2 ER diagram:

The Conventions in ER model are presented in Figure 1.

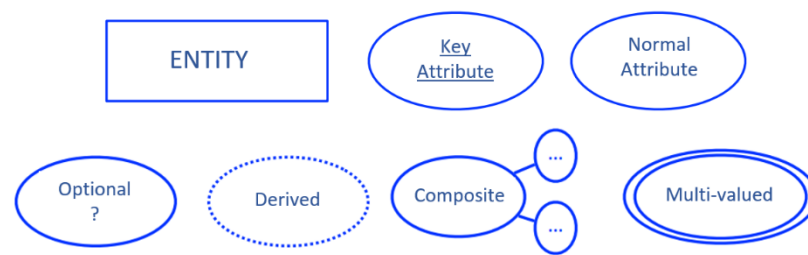


Figure 1. Conventions used in ER model diagram regarding entities and attributes.

In an ER model, entities are represented by uppercase letters inside rectangles. Key, normal, and optional attributes are written within an ellipse. Key attributes are underlined, and optional attributes are denoted with a question mark, written as "attribute ?". Derived attributes are also within an ellipse but have a dashed outline. Composite attributes are depicted with connections to their sub-attributes. Multi-value attributes are also represented within an ellipse, but with double outlines.

The conventions regarding relationships are presented in Figure 2.

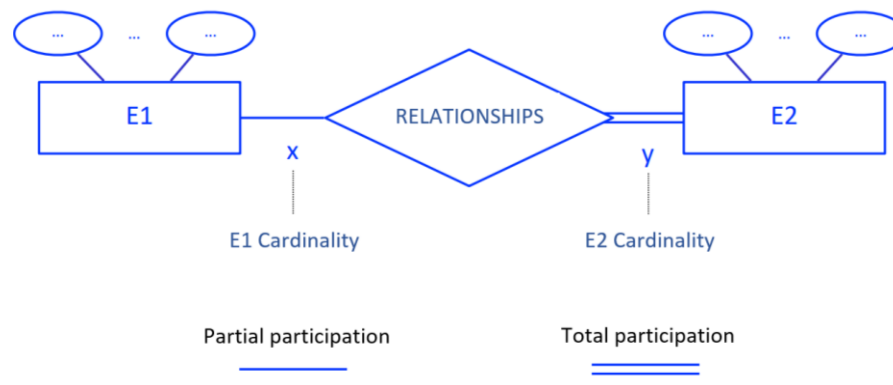


Figure 2. Conventions used in ER model diagram regarding relationships.

The relationships are represented inside diamonds. The attributes of the relationships are also represented inside ellipses.

4 | Relational model

As explained, the relational model is presented in the ZIP archive.

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Contrary to the ER model, in the relational model, attributes cannot be composite or multi-valued. Derived attributes are not mapped.

A table, denoted by $T(A_1, \dots, A_n)$ in the ER model, has the name T and attributes $A_1, \dots, A_n$. Analogously to the ER model, a key for the table is a minimal set of attributes that uniquely identify a record. There may be multiple possible keys, with the most suitable key for the given domain chosen as the primary key (which is underlined). The same primary keys in the relational model were used as the key attributes in the ER model.

The multi-valued attributes are mapped to an auxiliary table with a foreign key referencing the primary key of the T (as is the case with ARTIST and SONG). The foreign key and the remaining attributes define the key of the auxiliary table (as, SongID and Artist in table ARTIST).

In the visual representation of a relational schema, references are depicted using a link (arrow or directed arc) between the foreign key and the primary key.

Primary keys are underlined, and no special notation is used for optional attributes in the relational model.

Composite attributes have been mapped as follows (Table 3):

Table 3. Explanation of the translation of composite attributes from the ER model to the relational mode.

ER Model - textual syntax	Relation Model
UserEngagement(Plays,Downloads)	UEPlays UEDownloads
StorageCapacity(Total,Free)	SCTotal SCFree

Mapping of relationships:

- CREATOR_OF(USER, PLAYLIST, CreationDate) has a cardinality of 1:N and total participation of PLAYLIST. Therefore, in the relational model, the relationship attribute (CreationDate) is mapped to PLAYLIST, and the primary key of USER is defined as a foreign key in PLAYLIST (CreatorID $\rightarrow$ USER.UserID);
- All other relationships have a cardinality of M:N: A cross-reference table was created specifying the REL. The primary key of each table is the "Primary key of T1" + "Primary key of T2" (both foreign keys) and the attributes were mapped to the REL table (as shown in Table 4);
- In the relationships 'PLAYED_ON' and 'CONNECTED_TO' the corresponding dates serve as primary keys. Within these relationships, three attributes are required to uniquely identify each instance in the database.

Table 4. Primary keys and foreign keys ($\rightarrow$) in the tables referring to the RELs that have a cardinality of M:N.

PLAYED_ON	LIKES_THE	BELONGS_TO	SUBSCRIBE_TO	CONNECTED_TO
<u>PlaylistID</u> $\rightarrow$ PLAYLIST.PlaylistID	<u>UserID</u> $\rightarrow$ USER.UserID	<u>SongID</u> $\rightarrow$ SONG.SongID	<u>UserID</u> $\rightarrow$ USER.UserID	<u>UserID</u> $\rightarrow$ USER.UserID
<u>DeviceID</u> $\rightarrow$ DEVICE.DeviceID	<u>SongID</u> $\rightarrow$ SONG.SongID	<u>PlaylistID</u> $\rightarrow$ PLAYLIST.PlaylistID	<u>PlaylistID</u> $\rightarrow$ PLAYLIST.PlaylistID	<u>DeviceID</u> $\rightarrow$ DEVICE.DeviceID
<u>PlayDate</u>	LikeDate	-	SubsDate	<u>ConnectDate</u>